

A tangent-circle mean-value identity

for an arctangent integral and related log-sine integrals

$$\boxed{\int_0^\pi \arctan\left(\frac{\log(\sin x)}{x}\right) dx = -\pi \arctan \frac{2 \log 2}{\pi}.}$$

Overview. The arctangent integral is a consequence of a mean-value identity. The curve

$$w(x) = \log \sin x + ix$$

is the principal logarithm of a circle tangent to the origin. Therefore integrals of $F(w(x))$ are mean values of holomorphic functions on a circle, and the mean is the value at the center.

Throughout, Log and Arg denote the principal branches, and

$$\arctan \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

1. Evaluation of the arctangent integral

Put

$$u = \sin x e^{ix}, \quad 0 < x < \pi.$$

Since $\sin x > 0$, this has the immediate polar interpretation

$$|u| = \sin x, \quad \arg u = x. \tag{1}$$

Then, on the principal branch,

$$\text{Log } u = \log \sin x + ix.$$

To see where the circle comes from, write

$$u = X + iY = \sin x \cos x + i \sin^2 x.$$

Thus

$$X = \frac{1}{2} \sin 2x, \quad Y = \frac{1}{2}(1 - \cos 2x),$$

and therefore

$$X^2 + \left(Y - \frac{1}{2}\right)^2 = \frac{1}{4} \sin^2 2x + \frac{1}{4} \cos^2 2x = \frac{1}{4}.$$

So $u(x)$ lies on the circle with center $i/2$ and radius $1/2$. In complex notation this is

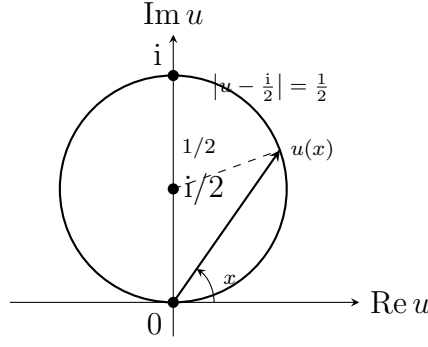
$$\left|u - \frac{i}{2}\right| = \frac{1}{2}. \tag{2}$$

This circle passes through 0 and i ; as x runs from 0 to π , the endpoints both limit to $u = 0$, and $x = \pi/2$ gives $u = i$. Hence $x \in (0, \pi)$ traverses the punctured boundary circle once.

The same fact has the compact one-line check

$$u - \frac{i}{2} = \sin x e^{ix} - \frac{i}{2} = \frac{e^{2ix}}{2i},$$

since $|e^{2ix}| = 1$.



Thus the curve $w(x) = \log \sin x + ix$ is the logarithmic image of a circle tangent to the origin.

Let

$$\Phi(u) = \text{Log Log } u,$$

with both logarithms principal. The only boundary singularity is at the tangency point $u = 0$, and its contribution vanishes; the rigorous punctured-disk check is given in Section 2. Thus the mean-value principle on the tangent circle¹ gives

$$\boxed{\int_0^\pi \text{Log Log}(\sin x e^{ix}) dx = \pi \text{Log Log } \frac{i}{2}.} \quad (3)$$

Now $\text{Log } u = \log \sin x + ix$ lies in the upper half-plane. Since $\log \sin x \leq 0$,

$$\arctan \frac{\log \sin x}{x} = \frac{\pi}{2} - \Im \text{Log}(\log \sin x + ix) = \frac{\pi}{2} - \Im \text{Log Log } u. \quad (4)$$

Therefore

$$\begin{aligned} I &= \int_0^\pi \arctan \frac{\log \sin x}{x} dx \\ &= \frac{\pi^2}{2} - \pi \Im \text{Log Log } \frac{i}{2}. \end{aligned}$$

But

$$\text{Log } \frac{i}{2} = -\log 2 + \frac{\pi i}{2},$$

so

$$\Im \text{Log Log } \frac{i}{2} = \text{Arg} \left(-\log 2 + \frac{\pi i}{2} \right) = \frac{\pi}{2} + \arctan \frac{2 \log 2}{\pi}.$$

Substitution gives

$$I = -\pi \arctan \frac{2 \log 2}{\pi}.$$

2. The hidden domain and the mean-value theorem

Let

$$D = \left\{ u : \left| u - \frac{i}{2} \right| < \frac{1}{2} \right\}.$$

The disk D lies in the upper half-plane and touches the origin only on the boundary. Hence the principal $\text{Log } u$ is holomorphic in D . Define

$$\Omega = \text{Log } D.$$

If $u = re^{i\theta}$, then $u \in D$ is equivalent to

$$0 < \theta < \pi, \quad 0 < r < \sin \theta.$$

¹If this circle mean-value principle is not familiar, see the Appendix for the polynomial and power-series explanation.

Thus, for $w = s + i\theta = \text{Log } u$,

$$\Omega = \{s + i\theta : 0 < \theta < \pi, \quad s < \log \sin \theta\}. \quad (5)$$

The boundary curve of Ω is exactly

$$w(x) = \log \sin x + ix, \quad 0 < x < \pi,$$

and the center of the tangent circle maps to

$$w_0 = \text{Log } \frac{i}{2} = -\log 2 + \frac{\pi i}{2}. \quad (6)$$

Mean-value theorem for the boundary curve. Let F be holomorphic in Ω , and set

$$G(u) = F(\text{Log } u), \quad u \in D.$$

Assume that G extends continuously to $\overline{D} \setminus \{0\}$, that $x \mapsto F(w(x))$ belongs to $L^1(0, \pi)$, and that the small arc at the tangency point does not contribute:

$$\delta \sup_{\substack{|u|=\delta \\ |u-i/2|\leq 1/2}} |F(\text{Log } u)| \longrightarrow 0, \quad \delta \rightarrow 0+. \quad (7)$$

Then

$$\int_0^\pi F(\log \sin x + ix) \, dx = \pi F\left(-\log 2 + \frac{\pi i}{2}\right). \quad (8)$$

Why a circle mean gives the center. For a polynomial this is only the fact that the non-constant Fourier modes average to zero. Indeed, if $G(z) = z^2$, then

$$G(c + re^{i\theta}) = c^2 + 2cre^{i\theta} + r^2e^{2i\theta},$$

and the averages of $e^{i\theta}$ and $e^{2i\theta}$ over a full circle vanish. Power series give the same rule for holomorphic functions; see the Appendix for this standard mean-value property. The Cauchy argument below is the rigorous form needed here, because our circle is tangent to the logarithmic singularity at 0.

Proof. For $0 < \delta < 1/2$, put

$$D_\delta = D \cap \{|u| > \delta\}.$$

The function G is holomorphic in D_δ and continuous on its closure. Cauchy's formula for this piecewise smooth Jordan domain gives

$$\frac{1}{2\pi i} \int_{\partial D_\delta} \frac{G(u)}{u - i/2} \, du = G\left(\frac{i}{2}\right). \quad (9)$$

The boundary ∂D_δ consists of the large circle $|u - i/2| = 1/2$, with small pieces around 0 removed, and a small arc of $|u| = \delta$. The omitted pieces of the large circle tend to zero by the assumed integrability of $x \mapsto F(\log \sin x + ix)$. The inner arc $|u| = \delta$ tends to zero by (7): its length is $O(\delta)$, while $u - i/2$ is bounded away from 0. Letting $\delta \rightarrow 0+$, only the tangent circle remains.

Finally parameterize it by

$$u = \frac{i}{2} + \frac{1}{2}e^{i\theta}, \quad \theta = 2x - \frac{\pi}{2}, \quad d\theta = 2 \, dx.$$

Equivalently,

$$du = 2i \left(u - \frac{i}{2}\right) \, dx,$$

and therefore

$$\frac{1}{2\pi i} \int_{|u-i/2|=1/2} \frac{G(u)}{u - i/2} \, du = \frac{1}{\pi} \int_0^\pi G(u(x)) \, dx.$$

Together with (9), this is exactly (8).

For the function used in Section 1, $F(w) = \text{Log } w$, the hypotheses can be checked explicitly. On the inner arc $|u| = \delta$, write $\text{Log } u = \log \delta + i\theta$, with $0 \leq \theta \leq \pi$. Then

$$|\text{Log}(\text{Log } u)| = O(1 + \log |\log \delta|),$$

so the left-hand side of (7) is $O(\delta \log |\log \delta|) \rightarrow 0$. Moreover,

$$|\text{Log } w(x)| = O(1 + \log |\log x|) \quad (x \rightarrow 0+),$$

and

$$|\text{Log } w(x)| = O(1 + \log |\log(\pi - x)|) \quad (x \rightarrow \pi-),$$

and both bounds are integrable. Thus (3) satisfies all the hypotheses of the theorem.

3. Interpretation of the theorem

The functional

$$\mathcal{T}[F] = \int_0^\pi F(\log \sin x + ix) dx$$

is evaluation at one point:

$$\boxed{\mathcal{T}[F] = \pi F(w_0), \quad w_0 = -\log 2 + \frac{\pi i}{2}.} \quad (10)$$

Geometrically, $w_0 = \text{Log}(i/2)$ is the logarithm of the center of the tangent circle. Formula (8) is the circle mean-value property transported by the map $u \mapsto \text{Log } u$.

4. Consequences

Choice of F	Identity	Reading
$F(w) = w^n$	$\int_0^\pi w(x)^n dx = \pi w_0^n$	Moments of the log-sine curve are determined by the center.
$F(w) = e^{tw}$, $\Re t > -1$	$\int_0^\pi (\sin x)^t e^{itx} dx = \pi 2^{-t} e^{\pi it/2}$	The diagonal beta-Fourier integral is a consequence of the same mean-value identity.
$F(w) = \text{Log } w$	$\int_0^\pi \text{Log } w(x) dx = \pi \text{Log } w_0$	The imaginary part gives the argument integral; Section 7 converts it to the arctangent.
$F(w) = \text{Log}(w + a)$, $a \in \mathbb{R}$	$\int_0^\pi \text{Log}(w(x) + a) dx = \pi \text{Log}(w_0 + a)$	Shifted arctangent and logarithmic quadratic identities.
$F(w) = 1/(w - a)$, with no pole in Ω or on the boundary	$\int_0^\pi \frac{dx}{w(x) - a} = \frac{\pi}{w_0 - a}$	The rational integrals used in parameter-differentiation solutions.
Meromorphic F , see Section 9	$\mathcal{T}[F] = \pi F(w_0) + \pi \sum r_j \frac{e^{a_j}}{e^{a_j} - i/2}$	Poles inside Ω contribute explicit residue terms.

Here and below

$$w(x) = \log \sin x + ix, \quad w_0 = -\log 2 + \frac{\pi i}{2}.$$

5. Centered symmetry

The cleanest form is obtained after subtracting the center:

$$Z(x) = w(x) - w_0 = \log(2 \sin x) + i \left(x - \frac{\pi}{2} \right).$$

Taking $F(w) = e^{t(w-w_0)}$ in (8), for $\Re t > -1$, gives

$$\boxed{\int_0^\pi e^{tZ(x)} dx = \pi.} \quad (11)$$

Differentiating at $t = 0$,

$$\boxed{\int_0^\pi \left[\log(2 \sin x) + i \left(x - \frac{\pi}{2} \right) \right]^n dx = 0, \quad n \geq 1.} \quad (12)$$

Thus all centered analytic moments vanish. For example, $n = 2$ yields

$$\int_0^\pi \log^2(2 \sin x) dx = \int_0^\pi \left(x - \frac{\pi}{2} \right)^2 dx = \frac{\pi^3}{12},$$

and

$$\int_0^\pi \log(2 \sin x) \left(x - \frac{\pi}{2} \right) dx = 0.$$

6. Moments and the beta-Fourier diagonal

For every integer $n \geq 0$, taking $F(w) = w^n$ in (8) gives

$$\boxed{\int_0^\pi (\log \sin x + ix)^n dx = \pi \left(-\log 2 + \frac{\pi i}{2} \right)^n.} \quad (13)$$

The case $n = 1$ recovers

$$\int_0^\pi \log \sin x dx = -\pi \log 2, \quad \int_0^\pi x dx = \frac{\pi^2}{2}.$$

The case $n = 2$ gives

$$\int_0^\pi x \log \sin x dx = -\frac{\pi^2}{2} \log 2$$

and

$$\int_0^\pi (\log \sin x)^2 dx = \pi (\log 2)^2 + \frac{\pi^3}{12}.$$

With $F(w) = e^{tw}$, $\Re t > -1$, one obtains

$$\boxed{\int_0^\pi (\sin x)^t e^{itx} dx = \frac{\pi}{2^t} e^{\pi it/2}.} \quad (14)$$

For real $t > -1$, real and imaginary parts give

$$\begin{aligned} \int_0^\pi (\sin x)^t \cos(tx) dx &= \frac{\pi}{2^t} \cos \frac{\pi t}{2}, \\ \int_0^\pi (\sin x)^t \sin(tx) dx &= \frac{\pi}{2^t} \sin \frac{\pi t}{2}. \end{aligned}$$

7. Logarithms, arctangents, and quadratic forms

Taking $F(w) = \text{Log } w$ gives

$$\boxed{\int_0^\pi \text{Log}(\log \sin x + ix) \, dx = \pi \text{Log}\left(-\log 2 + \frac{\pi i}{2}\right).} \quad (15)$$

Twice the real part gives

$$\boxed{\int_0^\pi \log((\log \sin x)^2 + x^2) \, dx = \pi \log\left((\log 2)^2 + \frac{\pi^2}{4}\right).} \quad (16)$$

The imaginary part gives the argument integral. Using

$$\arctan \frac{\log \sin x}{x} = \frac{\pi}{2} - \text{Arg}(\log \sin x + ix),$$

it is exactly equivalent to the arctangent integral from Section 1.

Here is the elementary triangle behind that identity. Put

$$\rho = -\log \sin x \geq 0, \quad w = \log \sin x + ix = -\rho + ix.$$

Then w lies in the closed second quadrant; at $x = \pi/2$, it lies on the positive imaginary axis. The angle between the positive imaginary axis and the vector w has tangent ρ/x , so

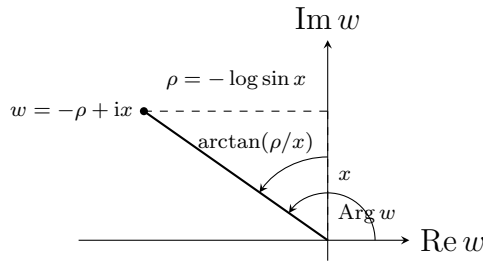
$$\text{Arg } w = \frac{\pi}{2} + \arctan \frac{\rho}{x}.$$

Since

$$\arctan \frac{\log \sin x}{x} = -\arctan \frac{\rho}{x},$$

we get

$$\arctan \frac{\log \sin x}{x} = \frac{\pi}{2} - \text{Arg } w.$$



More generally, for $\alpha > 0$,

$$I(\alpha) = \int_0^\pi \arctan \frac{\log(\alpha \sin x)}{x} \, dx$$

is obtained from $F(w) = \text{Log}(w + \log \alpha)$. Since for $x > 0$ and real A

$$\arctan \frac{A}{x} = \frac{\pi}{2} - \text{Arg}(A + ix), \quad \arctan \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \text{Arg}(A + ix) \in (0, \pi), \quad (17)$$

there is no branch ambiguity. The result is

$$\boxed{I(\alpha) = -\pi \arctan \frac{2\log(2/\alpha)}{\pi}, \quad \alpha > 0.} \quad (18)$$

Similarly, for real a , taking $F(w) = \text{Log}(w - a)$ and then taking twice the real part gives

$$\boxed{\int_0^\pi \log((\log \sin x - a)^2 + x^2) \, dx = \pi \log\left((\log 2 + a)^2 + \frac{\pi^2}{4}\right).} \quad (19)$$

8. Rational integrals

For real a , $F(w) = 1/(w - a)$ is holomorphic in Ω , since Ω lies strictly between the horizontal lines $\Im w = 0$ and $\Im w = \pi$. Hence

$$\boxed{\int_0^\pi \frac{dx}{\log \sin x - a + ix} = \frac{\pi}{-\log 2 - a + \pi i/2}}. \quad (20)$$

Separating real and imaginary parts,

$$\boxed{\int_0^\pi \frac{\log \sin x - a}{(\log \sin x - a)^2 + x^2} dx = \pi \frac{-\log 2 - a}{(-\log 2 - a)^2 + \pi^2/4}}, \quad (21)$$

$$\boxed{\int_0^\pi \frac{x}{(\log \sin x - a)^2 + x^2} dx = \pi \frac{\pi/2}{(-\log 2 - a)^2 + \pi^2/4}}. \quad (22)$$

Putting $a = -\log \alpha$, $\alpha > 0$, gives the rational integral usually found by differentiating a parameter:

$$\boxed{\int_0^\pi \frac{x}{x^2 + \log^2(\alpha \sin x)} dx = \frac{\pi^2/2}{\log^2(2/\alpha) + \pi^2/4}}. \quad (23)$$

Higher powers are just as direct: if $m \geq 1$ is an integer and $a \notin \overline{\Omega}$, then

$$\boxed{\int_0^\pi \frac{dx}{(\log \sin x - a + ix)^m} = \frac{\pi}{(-\log 2 - a + \pi i/2)^m}}. \quad (24)$$

9. Meromorphic extension

The holomorphic theorem also has a useful residue version. Suppose F is meromorphic in Ω , holomorphic at w_0 , and has finitely many simple poles $a_j \in \Omega$, with residues

$$r_j = \operatorname{Res}_{w=a_j} F(w).$$

Assume no pole lies on the boundary curve, that $G(u) = F(\operatorname{Log} u)$ has continuous boundary values on $\partial D \setminus \{0\}$, that $x \mapsto F(w(x))$ belongs to $L^1(0, \pi)$, and that the tangency condition (7) holds for F . Then

$$\boxed{\int_0^\pi F(\log \sin x + ix) dx = \pi F(w_0) + \pi \sum_{a_j \in \Omega} r_j \frac{e^{a_j}}{e^{a_j} - i/2}}. \quad (25)$$

Indeed, in the u -plane one applies the residue theorem to

$$\frac{F(\operatorname{Log} u)}{u - i/2}.$$

A pole a_j in the w -plane corresponds to $u_j = e^{a_j}$. Since

$$\operatorname{Log} u - a_j = \frac{u - u_j}{u_j} + O((u - u_j)^2),$$

the residue of $F(\operatorname{Log} u)$ at u_j is $r_j u_j$, producing the factor

$$\frac{u_j}{u_j - i/2} = \frac{e^{a_j}}{e^{a_j} - i/2}.$$

10. Relation to parameter methods

A common real-variable solution introduces a parameter, differentiates an arctangent integral, evaluates a rational integral, and then integrates the parameter back. In this notation those steps are simply different choices of F in the same functional:

$$F(w) = \text{Log}(w + \log \alpha) \quad \text{and} \quad F(w) = \frac{1}{w - a}.$$

Thus the parameter method recovers particular consequences of the mean-value identity. Formula (8) gives the corresponding functional identity directly.

Appendix. The circle mean-value property

This appendix recalls the standard mean-value property used in the proof. It is included only to make explicit the elementary mechanism behind the circle average; the derivation of the tangent circle itself is part of Section 1.

Let

$$z = c + re^{i\theta}$$

be a circle with center c . If $G(z) = z$, then

$$G(c + re^{i\theta}) = c + re^{i\theta}.$$

The average of the constant term is c , and

$$\frac{1}{2\pi} \int_0^{2\pi} e^{i\theta} d\theta = 0.$$

Therefore the average is $c = G(c)$.

If $G(z) = z^2$, then

$$(c + re^{i\theta})^2 = c^2 + 2cre^{i\theta} + r^2e^{2i\theta}.$$

Again the rotating terms average to zero:

$$\frac{1}{2\pi} \int_0^{2\pi} e^{in\theta} d\theta = 0, \quad n \geq 1.$$

Only $c^2 = G(c)$ remains. The same argument works for every polynomial. If G is holomorphic on a neighborhood of the closed disk $|z - c| \leq r$, its power series at c converges on the circle, and the same averaging argument gives

$$\frac{1}{2\pi} \int_0^{2\pi} G(c + re^{i\theta}) d\theta = G(c).$$

This is the circle mean-value property. In the body of the note we use Cauchy's formula to handle one extra issue: our circle is tangent to 0, where $\text{Log } u$ has a boundary singularity.